

PRACTICAL EXAM TRAINER WORKSHEET WATER & PREVENTIVE MAINTENANCE FOUNDATION COFFEE TECHNICIANS PROGRAM

实作测验 培训师工作表

水与预防性维护基础课程

咖啡技师学程

This optional trainer worksheet is designed to help a trainer efficiently proctor, observe, and score the practical exam for up to six (6) learners based on the instructions within the Trainer Guide. Trainers may also use other methods to record the individual points for the different tasks included in the exam. Following the completion of the practical exams, trainers will need to transfer point totals to the appropriate spaces on the Practical Exam Learner Worksheet for each student.

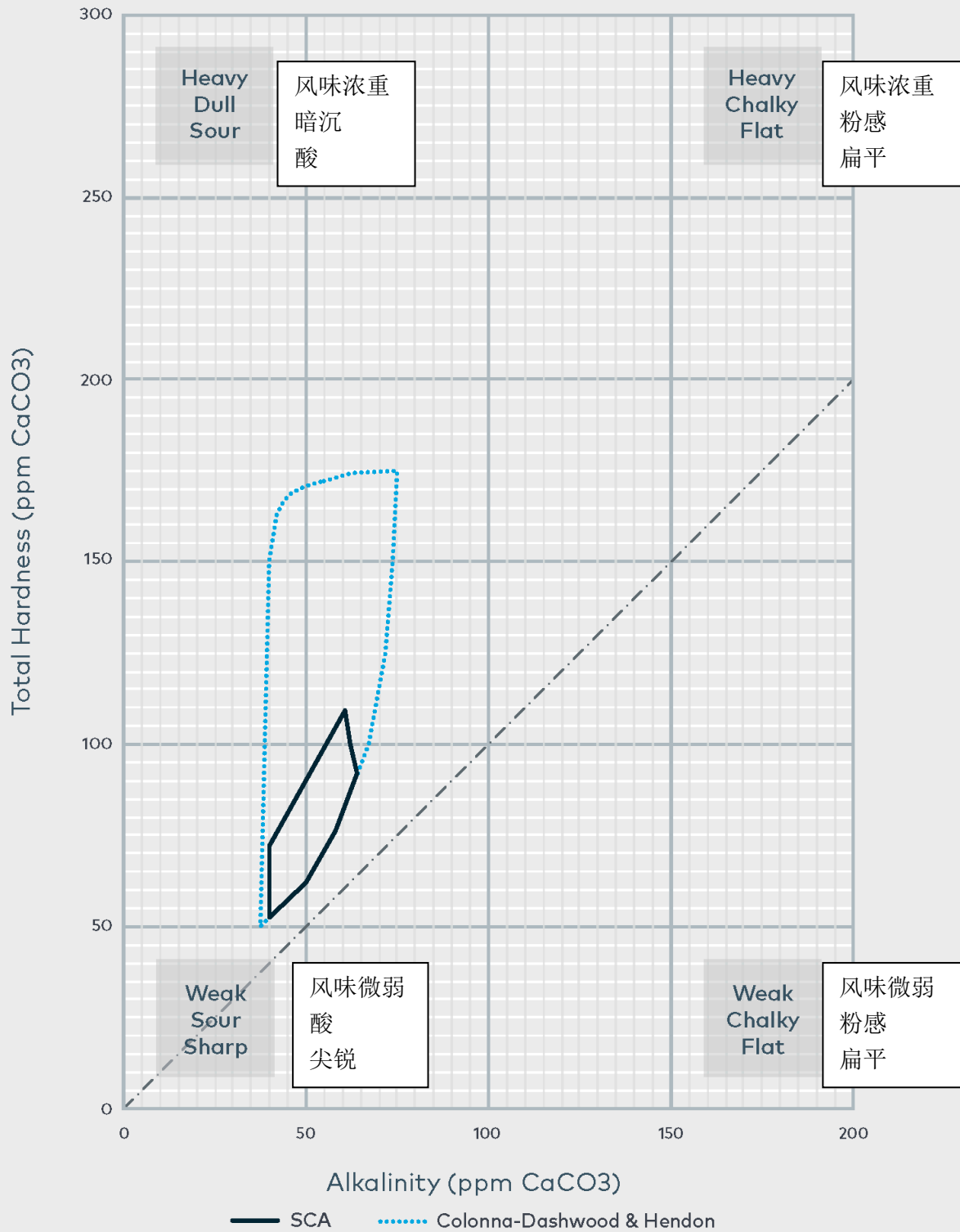
本工作表（非必要）的设计旨在协助培训师透过有效率的方式，为最多六（6）名学员监考、观察和评量实作测验。培训师也可透过其他方式，记录学员在测验中各项任务的分数。完成施测后，培训师应将成绩加总，并妥善注记在每个学员的『实作测验学员工作表』上。

施测日期 Date of Course: _____

LEARNERS 学习者:

1.
2.
3.
4.
5.
6.

Use this chart as a answer key for Section 1, Skill B. 使用此图表作为第一部分、技能 B 之解答参考。



Section 1 – Measuring Water Composition

第一部分 – 水中成分量测

Skill A - Measure the Total Hardness and Alkalinity of a water sample

技能 A – 水样总硬度与碱度之量测

Task	Pts						
Learner properly collects samples and fills testing vials 学习者正确地搜集样本并填充试瓶。	1						
Learner uses appropriate technique for adding reagent drops and agitating sample (1pt per vial) 学习者使用适当的技巧，于水样中添加试剂并进行搅拌。(每个试瓶 1 分)	2						
Learner correctly recognizes color change and drops added for Total Hardness 学员正确地辨认测量总硬度时的颜色变化及添加的滴定量。	1						
Learner correctly recognizes color change and drops added for Alkalinity 学员正确地辨认测量碱度时的颜色变化及添加的滴定量。	1						
Learner correctly records measurements on Learner Worksheet 学员正确地将测量结果记录在学员工作表。	1						
Total	6						

Skill B - Record Total Hardness and Alkalinity measurements on the SCA Water Chart

技能 B-在 SCA 水质图记录总硬度及碱度测量值

Task	Pts						
Learner correctly performs calculation to convert measurements into ppm CaCO ₃ (1pt per value) 学员正确地执行运算将测量结果转换成 ppm CaCO ₃ 值。(每个数值一分)	2						
Learner accurately plots the Total Hardness result obtained on the provided chart 学员准确地在提供的图表中标示出所获得的总硬度结果。	2						
Learner accurately plots the Alkalinity result obtained on the provided chart	2						

学员准确地在提供的图表中标示出所获得的碱度结果。							
Total	6						

Skill C - Use an Electrical Conductivity meter to measure the estimated TDS of a water sample

技能 C-使用导电度计测量水质样本的 TDS 概略值。

Task 任务	Pts 分数						
Learner properly collects water sample 学员正确地收集水质样本。	1						
Learner uses appropriate technique for using electrical conductivity meter 学员使用适当的技巧操作导电度计。	1						
Learner correctly records estimated TDS measurement on Learner Worksheet 学员正确地将 TDS 测量结果记录在学员工作表。	1						
Total 总分	3						

Trainer Comments: 讲师评语

Section 2 - Water Treatment System Maintenance

Replace a cartridge in a drop-in style water filter system

更换置入式过滤系统的滤芯

Task	Pts						
Learner turns off water supply to water filter 学员关闭滤水器的供水。	2						
Learner removes filter housing and old water filter cartridge 学员拆下滤水器外壳并移除老旧滤芯。	2						
Learner properly cleans and sanitizes filter housing 学员正确地清洁并消毒滤水器外壳。	2						
Learner installs new water filter cartridge and reassembles housing properly 学员安装新的滤芯并正确地重新组装外壳。	2						
Learner turns on water supply to filter and flushes system appropriately 学员打开滤水器的供水并适当地冲洗过滤系统。	2						
Filter system does not leak 过滤系统不会漏水。	2						
Learner records last change date and replacement on Learner Worksheet 学员在学员工作表上记录最后更换日期及更换项目。	1						
Total 总分	13						

Trainer Comments:

Section 3 - Coffee Brewer and Espresso Grinder Preventive Maintenance

第三章-咖啡冲煮机及义式磨豆机预防性维护

Skill A - Clean a coffee batch brewer level probe

技能 A- 清洁滴滤式咖啡机的水位探针

Task	Pts						
Learner confirms that coffee brewer is disconnected from electrical power 学员确认滴滤式咖啡机未接上电源。	2						
Learner removes required access panel and places screws in a secure location 学员拆除所需的检修孔盖并将螺丝放置在妥善的位置。	2						
Learner removes level probe without damaging probe or mounting bushings 学员在不损害探针或安装套管的情况下拆除水位探针。	2						
Learner uses appropriate means to fully clean the level probe 学员使用适当的方法完整地清洁水位探针。	2						
Learner reinstalls level probe and connects wires if required 学员重新安装水位探针并根据需求接上电线。	2						
Learner reassembles access panel and tightens screws appropriately 学员重新安装检修孔盖并适当地拧紧螺丝。	2						
Learner records activity completion on Learner Worksheet 学员将活动完成情形记录在学员工作表上。	1						
Total	13						

Skill B - Replace the burrs in an espresso grinder

技能 3B-更换义式磨豆机的刀盘

Task	Pts						
Learner unplugs espresso grinder before beginning service 学员在开始维护前拔除义式磨豆机的插头。	2						
Learner opens grinding chamber and removes worn grinding burrs 学员拆开研磨腔室并移除磨损的刀盘。	3						
Learner thoroughly cleans grinding chamber and burr mounting surfaces 学员彻底地清洁研磨腔室及刀盘的安装面。	2						
Learner installs new grinder burrs, using proper screw tightening sequence 学员使用合适的螺丝拴紧方式安装新的刀盘。	3						
Learner reassembles grinder and connects to electrical outlet 学员重新安装磨豆机并接上插座。	3						
Learner tests for proper function and adjusts to approximately correct setting 学员测试功能是否正常并调整至大略正确的设定。	2						
Learner records activity completion on Learner Worksheet 学员将活动完成情形记录在学员工作表上。	1						
Total	16						

Trainer Comments:

Section 4 - Espresso Machine Preventive Maintenance 义式咖啡机的预防性维护

Skill A - Replace an espresso machine group screen and gasket

技能 A- 更换义式咖啡机的冲煮头分水网和垫圈

Task	Pts						
Learner properly removes group screen and screw 学员正确地拆除冲煮头分水网及螺丝。	2						
Learner properly removes group gasket without damaging brewing group 学员正确地拆除垫圈且未损坏冲煮头。	3						
Learner cleans group gasket channel and group 学员将垫圈沟槽及冲煮头清洁干净。	2						
Learner orients group gasket correctly when installing 学员以正确的一面定位垫圈。	2						
Learner properly installs new group gasket 学员正确地安装新垫圈。	2						
Learner installs group screen and screw correctly, using appropriate screw torque 学员正确地安装冲煮头分水网及螺丝，并使用适当的力道拧紧螺丝。	2						
Learner records activity completion on Learner Worksheet 学员将活动完成情形记录在学员工作表上。	1						
Total	14						

Skill B - Rebuild an espresso machine steam wand joint

技能 B-重新组装一只义式咖啡机的蒸气棒接头

Task	Pts						
Learner properly removes steam wand from steam valve 学员正确地蒸气阀上拆下蒸气棒。	2						
Learner properly removes wear parts as directed without damage to equipment 学员根据引导正确地拆除磨损零件且未对设备造成损坏。	2						
Learner cleans steam wand joint area and valve as directed	2						

学员根据引导将蒸气棒接头区块及蒸汽阀清洁干净。							
Learner properly installs replacement parts, applying lubricant where applicable 学员正确地安装更换零件，并在适当的地方添加润滑剂。	3						
Learner properly reassembles steam wand to steam valve 学员正确地将蒸气棒重新安装回蒸气阀。	2						
Steam wand joint does not leak when tested 确认测试时蒸气棒接头不会泄漏。	2						
Learner records activity completion on Learner Worksheet 学员将活动完成情形记录在学员工作表上。	1						
Total 总分	14						

Skill C - Complete an espresso machine inspection

技能 C-完成义式咖啡机的检查

Task	Pts 分数						
Learner correctly records steam boiler pressure on Learner Worksheet 学员正确地在学员工作表上记录蒸气锅炉压力。	1						
Learner observes pump pressure using blank portafilter basket or thermofilter device 学员透过使用无孔滤杯或测温测压把手观察泵浦压力。	2						
Learner correctly records pump pressure on Learner Worksheet 学员正确地将蒸气泵浦压力记录在学员工作表上。	1						
Learner tests anti-vacuum valve and records operating state on Learner Worksheet 学员测试真空排除阀并将运作状况记录在学员工作表上。	2						

Learner inspects safety valve and records operating state on Learner Worksheet 学员检查安全阀并将运作状况记录在学员工作表上。	1						
Learner tests drain function and records operating state on Learner Worksheet 学员测试排水功能并将运作状况记录在学员工作表上。	2						
Learner inspects machine for water leaks and records operating state on Learner Worksheet 学员检查机器是否漏水并将运作状况记录在学员工作表上。	2						
Learner inspects machine for steam leaks and records operating state on Learner Worksheet 学员检查机器的蒸气是否会泄漏并将状况记录在学员工作表上。	2						
Learner measures group water debit and records observations on Learner Worksheet 学员测量冲煮头出水量，并将观测结果记录于学员工作表	2						
Total 总分	15						

Trainer Comments: 讲师评语

Use this copy of the Preventive Maintenance Checklist as an answer key for the exam.

Practical Exam - Preventive Maintenance Checklist

实务测验-预防性维护核对表

Water Filtration System 水过滤系统		Service Required? 需要维护?	Service Complete 维护完成
Service or Inspection Item 维护或检查项目	Observation 观察结果		
Measure Total Hardness 测量总硬度		Yes / No 是/否	
Measure Alkalinity 测量碱度		Yes / No 是/否	
Measure TDS 测量 TDS		Yes / No 是/否	
Service water filter cartridge (record last changed date) 更换滤水器滤芯(记录最后更换日期)			
Traditional Espresso Machine 传统义式咖啡机		Service Required? 需要维护?	Service Complete 维护完成
Service or Inspection Item 维护或检查项目	Observation 观察结果		
Clean brewing group 清洁冲煮头			
Replace group gaskets 更换冲煮头垫圈			
Replace group screens 更换冲煮头分水网			
Rebuild steam wand joint 重新组装蒸气棒接头			
Inspect for steam valve and steam wand joint leaks 检查蒸气阀及蒸汽棒接头是否泄漏		Yes / No 是/否	
Record steam boiler pressure 记录蒸气锅炉压力		Yes / No 是/否	
Record pump pressure 记录泵浦压力		Yes / No 是/否	
Inspect steam boiler safety valve 检查蒸气锅炉安全阀		Yes / No 是/否	

Test anti-vacuum valve function 测试真空排除阀功能		Yes / No 是/否	
Inspect expansion valve 检查膨胀阀		Yes / No 是/否	
Test auto fill system function / steam boiler water level 测试自动补水系统功能/蒸气锅炉水位		Yes / No 是/否	
Test machine drain function 测试机器的排水功能		Yes / No 是/否	
Inspect for water leaks overall 彻底检查是否漏水		Yes / No 是/否	
Inspect for steam leaks overall 彻底检查蒸气是否泄漏		Yes / No 是/否	
Coffee Batch Brewer 滴滤式咖啡机		Service Required? 需要维护?	Service Complete 维护完成
Service or Inspection Item 维护或检查项目	Observation 观察结果		
Clean level probe (record any build-up) 清洁水位探针(记录任何结垢)			
Espresso Grinder 义式磨豆机		Service Required? 需要维护?	Service Complete 维护完成
Service or Inspection Item 维护或检查项目	Observation 观察结果		
Clean grinding chamber and replace burrs 清洁研磨腔室并更换刀盘			
Adjust grind setting 调整研磨设定			